Answer key. Answers are in magenta.

NRES 746 – Week 3, Lecture 2: A Probability Puzzle

Instructions: Work alone for about 5-8 minutes, guessing before you compute anything. Then compare with a neighbor.

Part 1: Sevens vs. doubles

Roll two fair six-sided dice (one red, one blue, so you can tell them apart).

1a. Before doing any counting: which do you think is more likely, rolling a **sum of 7**, or rolling **doubles** (both dice show the same number)? Circle one:

sum of 7 doubles equally likely

Equally likely (see 1c and 1d).

1b. Now check your guess. First: how many equally-likely outcomes are there in the full sample space Ω (all ordered pairs of red/blue results)?

$\|\Omega\| = 36$

1c. How many of those outcomes sum to 7? List them if it helps.

$P(\text{sum} = 7) = 6/36 = 1/6$ (1,6), (2,5), (3,4), (4,3), (5,2), (6,1)

1d. How many outcomes are doubles?

$P(\text{doubles}) = 6/36 = 1/6$

1e. Was your guess in 1a correct? If not, what made doubles or sevens feel more/less likely than it actually is?

Open-ended. They're tied at 6 of 36. (7 is the most common sum, so it ties doubles; any other sum loses.)

Part 2: How many tickets?

A small regional lottery asks you to pick 5 distinct numbers from 1 to 30 (order doesn't matter - {3, 8, 14, 20, 29} is the same ticket as {29, 20, 14, 8, 3}). How many different tickets are possible?

Leave your answer as an expression (use factorials!). No need to provide an actual integer value.

Hint: how many choices do you have for the first number you write down? How many for the second, given the first is already used? ... and then how do you account for the fact that order doesn't matter?

$\binom{30}{5} = \frac{30!}{5!\,25!} = \frac{30 \times 29 \times 28 \times 27 \times 26}{5!}$ (= 142,506)

Part 3: A teaser (guess only)

Suppose there are 25 people in this room. Do you think it's more likely than not (better than a 50% chance) that **at least two of them share a birthday**? Circle your guess:

yes no

We'll come back to this one once we have the tools to compute it exactly. You can give it a try, but it's more difficult than the lottery problem!

Yes, about 57%. Use the complement: person k must avoid the $k-1$ birthdays already taken, so

$$P(\text{no match}) = \frac{365}{365} \cdot \frac{364}{365} \cdots \frac{341}{365} = \prod_{k=1}^{25} \frac{366-k}{365} \approx 0.431, \qquad P(\text{at least one match}) \approx 0.569.$$

It feels unlikely because we compare 25 people with 365 days, but what matters is the number of pairs: $\binom{25}{2} = 300$ chances for a match.